

UK Patent Application - Claims

CLAIMS

1. A computer-implemented system for predictive communication outcome optimisation comprising: a processor; a memory storing instructions executable by the processor; an actor profile database storing actor representations, each actor representation including a plurality of communication variables quantified across five ontological layers comprising cognitive variables, emotional variables, sociocultural variables, behavioral variables, and strategic variables; an architect state module storing current skill levels across three dimensions comprising observation skill, adaptation skill, and self-awareness skill, and storing current levels for five confounding variables comprising stress level, fatigue level, emotional state, overconfidence level, and ego investment level; a confounder penalty engine configured to apply multiplicative penalty factors to base skill levels based on current confounder levels, wherein stress level reduces observation skill by thirty percent and adaptation skill by thirty percent, fatigue level reduces observation skill by twenty percent and adaptation skill by twenty percent, emotional state reduces observation skill by twenty percent and self-awareness by thirty percent, overconfidence reduces self-awareness by forty percent, and ego investment reduces adaptation skill by thirty percent; a strategy configuration interface receiving user input defining five strategy lever settings on bipolar scales from negative one to positive one comprising warmth, competence, dominance, status, and rapport; a Monte Carlo simulation engine configured to execute a plurality of iterations, each iteration sampling actor variable values and calculating predicted changes in three strategic outcomes comprising change in notoriety, change in respect, and change in wealth, wherein notoriety is calculated as mean across actors of product of awareness, understanding, and support willingness, respect is calculated as mean across actors of product of feels understood, consideration received, and value alignment perceived, and wealth comprises sum of direct financial gains, pipeline opportunity value, and executed project value; a strategic index calculation module configured to compute seven effectiveness metrics comprising ROI Index calculated as fifty plus fifty times weighted sum of outcome changes, Confidence Delta measuring estimate certainty change, Resistance Index measuring interaction friction, Energy Cost measuring cognitive and emotional load, Influence Depth measuring belief change achieved, Leverage Activation measuring strategic advantage creation, and Narrative Coherence measuring message clarity; a post-interaction recording interface configured to capture actual confounder levels experienced, actual strategy lever deployments, actual strategic index values observed, and actual outcome changes achieved; an accuracy calculation module configured to compute prediction error metrics by comparing predicted values to actual values across confounders, strategy levers, strategic indexes, and strategic outcomes; and a bias identification module configured to identify systematic overestimation or underestimation patterns for specific variables or actor types based on accumulated accuracy metrics across multiple interactions.

2. The system of claim 1 further comprising a cognitive journal module configured to log cognitive distortion episodes, each episode including an external trigger observation, a somatic feeling description, a core belief interpretation, and a positive reframe statement,

wherein each episode is linkable to a specific actor profile via a foreign key relationship enabling identification of actor types that systematically trigger cognitive distortions for the user.

3. The system of claim 2 wherein the cognitive journal module is further configured to map logged distortions to Jungian psychological polarities and generate integration experiment tasks encouraging exploration of opposite psychological poles, and wherein cognitive distortion logging frequency and successful reframing rate are aggregated into a self-awareness score that modifies the architect self-awareness skill level used in subsequent simulations.

4. The system of claim 1 further comprising a communication projects module configured to organize communication tasks into strategic projects, each project including a strategic objective definition, a set of linked actor profiles, a hierarchy of interaction tasks explicitly typed as communication events, pre-configured simulation templates for common scenarios within project type, and aggregated progress metrics tracking cumulative strategic outcome changes across all project interactions.

5. The system of claim 4 wherein the communication projects module includes specialized project types comprising stakeholder alignment projects with templates for buy-in communication sequences, conflict resolution projects with phased communication workflows, relationship building projects with systematic cultivation strategies, and negotiation projects with multi-round preparation templates.

6. The system of claim 1 wherein the five cognitive variables comprise analytical reasoning, abstract thinking, pattern recognition, knowledge depth, learning rate, cognitive flexibility, memory capacity, and processing speed; the seven emotional variables comprise emotional intelligence, empathy, stress resilience, emotional stability, optimism bias, emotional expressiveness, and conflict tolerance; the six sociocultural variables comprise cultural fluency, social status sensitivity, authority response patterns, group identity strength, formality preference, and network centrality; the seven behavioral variables comprise communication style, verbosity, interruption tendency, question frequency, risk tolerance, decision speed, and detail orientation; and the six strategic variables comprise goal clarity, strategic thinking, long-term orientation, negotiation skill, adaptability, and relationship motivation.

7. The system of claim 1 wherein the Monte Carlo simulation engine is configured to execute between one hundred and ten thousand iterations with default iteration count of one thousand, and wherein simulation execution time for one thousand iterations with thirty-four actor variables is approximately zero point four to zero point five seconds using vectorized NumPy operations.

8. The system of claim 1 wherein the confounder penalty engine applies penalties multiplicatively such that effective observation skill is calculated as base observation skill multiplied by quantity one minus stress level multiplied by zero point three multiplied by quantity one minus fatigue level multiplied by zero point two multiplied by quantity one minus emotional state multiplied by zero point two, effective adaptation skill is calculated as base adaptation skill multiplied by quantity one minus stress level multiplied by zero point three multiplied by quantity one minus fatigue level multiplied by zero point two multiplied by quantity one minus ego investment multiplied by zero point three, and effective self-awareness is calculated as base self-awareness multiplied by

quantity one minus quantity one minus emotional state multiplied by zero point three multiplied by quantity one minus overconfidence multiplied by zero point four.

9. The system of claim 1 wherein the strategic index calculation module computes ROI Index according to the formula ROI Index equals fifty plus fifty multiplied by open parenthesis weight_N multiplied by change in notoriety plus weight_R multiplied by change in respect plus weight_W multiplied by normalized change in wealth close parenthesis, where weights sum to one and are user-configurable based on current strategic priorities.

10. The system of claim 1 wherein the actor profile database stores each actor variable as a data structure including alpha parameter, beta parameter, observation count, and last updated timestamp, enabling representation of estimation uncertainty and Bayesian updating based on empirical observations.

11. A computer-implemented method for predictive communication outcome optimisation comprising the steps of: storing in a database a plurality of actor profiles, each profile including communication variables quantified across five ontological layers comprising cognitive, emotional, sociocultural, behavioral, and strategic dimensions; receiving architect state input defining current base skill levels for observation, adaptation, and self-awareness, and current confounder levels for stress, fatigue, emotional state, overconfidence, and ego investment; calculating effective skill levels by applying multiplicative penalty factors based on confounder levels, wherein stress reduces observation and adaptation by thirty percent, fatigue reduces observation and adaptation by twenty percent, negative emotional state reduces observation by twenty percent and self-awareness by thirty percent, overconfidence reduces self-awareness by forty percent, and ego investment reduces adaptation by thirty percent; receiving strategy configuration input defining five bipolar lever settings for warmth, competence, dominance, status, and rapport on scales from negative one to positive one; executing Monte Carlo simulation comprising a plurality of iterations, each iteration sampling actor variable values and calculating predicted changes in notoriety, respect, and wealth based on effective skills and strategy settings; computing seven strategic indexes comprising ROI Index as weighted outcome change, Confidence Delta measuring estimate certainty change, Resistance Index measuring friction, Energy Cost measuring load, Influence Depth measuring belief change, Leverage Activation measuring advantage creation, and Narrative Coherence measuring message clarity; presenting predicted outcome distributions with confidence intervals to user for interaction preparation; receiving post-interaction input capturing actual confounder levels, actual strategy deployments, actual strategic index values, and actual outcome changes; calculating accuracy metrics by comparing predicted values to actual values across all dimensions; identifying systematic prediction biases by analyzing error patterns across multiple interactions for specific variables or actor types; and storing accuracy metrics and bias patterns for display to user and incorporation into future simulation calibration.

12. The method of claim 11 further comprising logging cognitive distortion episodes including external trigger, somatic feeling, core belief, and positive reframe; linking each episode to a specific actor profile; mapping distortions to Jungian psychological polarities; generating integration experiment tasks; and aggregating distortion logging frequency and reframing success into self-awareness score modifying architect skill level.

13. The method of claim 11 further comprising organizing communication tasks into strategic projects; linking multiple actor profiles to each project; providing pre-configured simulation templates for project-specific scenarios; tracking cumulative strategic outcome changes across project interactions; and generating alerts when relationship degradation in individual interactions threatens project-level objectives.

14. The method of claim 11 wherein executing Monte Carlo simulation comprises sampling from Beta distributions for each actor variable using alpha and beta parameters, applying effective skill multipliers to outcome calculation functions, incorporating strategy lever settings as inputs to outcome functions, and aggregating results across iterations to produce mean outcome predictions and ninety-five percent confidence intervals.

15. The method of claim 11 wherein calculating effective skills applies penalties multiplicatively according to formulas effective observation equals base observation times one minus stress times zero point three times one minus fatigue times zero point two times one minus one minus emotional state times zero point two, effective adaptation equals base adaptation times one minus stress times zero point three times one minus fatigue times zero point two times one minus ego investment times zero point three, and effective self-awareness equals base self-awareness times one minus one minus emotional state times zero point three times one minus overconfidence times zero point four.

16. A non-transitory computer-readable storage medium storing instructions that when executed by a processor cause the processor to perform operations comprising: maintaining an actor profile database storing communication variables across five ontological layers for a plurality of actors; maintaining architect state including base skill levels and current confounder levels; applying multiplicative confounder penalties to calculate effective skill levels; receiving strategy lever configurations for warmth, competence, dominance, status, and rapport; executing Monte Carlo simulation to predict changes in notoriety, respect, and wealth; computing seven strategic effectiveness indexes; recording actual post-interaction outcomes; calculating prediction accuracy metrics; and identifying systematic bias patterns across interactions.

17. The computer-readable medium of claim 16 wherein the operations further comprise logging cognitive distortions with actor linkage, mapping to Jungian polarities, generating integration tasks, and incorporating self-awareness metrics into skill calculations.

18. The computer-readable medium of claim 16 wherein the operations further comprise organizing communication tasks into strategic projects with linked actors, scenario templates, and cumulative outcome tracking.

19. The computer-readable medium of claim 16 wherein executing Monte Carlo simulation comprises sampling from Beta distributions, applying multiplicative confounder penalties according to stress reducing observation and adaptation by thirty percent, fatigue reducing observation and adaptation by twenty percent, emotional state reducing observation by twenty percent and self-awareness by thirty percent, overconfidence reducing self-awareness by forty percent, and ego investment reducing adaptation by thirty percent, incorporating strategy lever settings, and aggregating across iterations.

20. The computer-readable medium of claim 16 wherein computing strategic effectiveness indexes comprises calculating ROI Index as fifty plus fifty times weighted sum of normalized outcome changes, Confidence Delta as change in variable estimation certainty, Resistance Index as aggregated friction signals, Energy Cost as average of

cognitive and emotional load, Influence Depth as degree of belief change, Leverage Activation as strategic advantage creation, and Narrative Coherence as message clarity measurement.

21. The system of claim 1 wherein the notoriety outcome is calculated as one divided by n multiplied by summation over i from one to n of awareness of actor i multiplied by understanding of actor i multiplied by support willingness of actor i , the respect outcome is calculated as one divided by n multiplied by summation over i from one to n of feels understood by actor i multiplied by consideration received from actor i multiplied by value alignment perceived by actor i , and the wealth outcome is calculated as sum of direct financial gains plus pipeline opportunity value weighted by execution probabilities plus executed project value realizations.

22. The system of claim 1 wherein the actor profile database implementation comprises a PostgreSQL relational database with JSONB columns storing variable structures, each variable structure including alpha parameter for Beta distribution shape, beta parameter for Beta distribution shape, observation count tracking empirical updates, and last updated timestamp.

23. The system of claim 1 wherein the post-interaction recording interface comprises a modal dialog component with four sections: actual confounders section with five sliders for stress, fatigue, emotional state, overconfidence, and ego investment ranging from zero to one hundred; actual strategy levers section with five bipolar sliders for warmth, competence, dominance, status, and rapport ranging from negative one hundred to positive one hundred; actual strategic indexes section with seven assessment inputs; and actual outcomes section capturing changes in notoriety, respect, and wealth plus interaction duration and free-text notes.

24. The method of claim 11 further comprising the step of presenting accuracy trend visualizations showing prediction error reduction over time, enabling quantification of architect's improving predictive capability through accumulated interaction experience and empirical validation.

25. The method of claim 11 wherein identifying systematic prediction biases comprises analyzing whether architect consistently overestimates effective adaptation skill under stress conditions, whether outcomes consistently underperform predictions when interacting with high-status actors, whether specific actor archetypes trigger estimation errors, and generating recommendations for bias mitigation including recalibration of base skill estimates or increased attention to specific confounder severity.

26. A computer-implemented training progress tracking system for simulation-based learning comprising: a processor; a memory storing instructions executable by the processor; a training metrics database storing five performance indicators per user comprising total interactions count, successful ROI count, success rate percentage, adaptive pivots count, and biases recognized count; a metric collection engine configured to automatically increment total interactions on simulation completion, increment successful ROI when actual ROI $\geq$ target ROI, calculate success rate as $(\text{successful ROI} / \text{total interactions}) \times 100$, increment adaptive pivots when $|\text{ROI_actual} - \text{ROI_target}| > \text{threshold}$ AND strategy changes, and increment biases recognized when confirmation bias, anchoring bias, or halo effect detected; a real-time update mechanism comprising database event triggers on metric changes, PostgreSQL LISTEN/NOTIFY for event

propagation, and WebSocket connections transmitting JSON payloads to clients without polling; and a dashboard visualization component displaying five metric cards in responsive grid layout with tooltips, color-coded feedback, and animated transitions on value changes.

27. The system of claim 26 wherein the adaptive pivots counter employs pivot threshold θ inversely proportional to adaptation skill level, such that higher adaptation skill requires larger ROI gaps to trigger pivot detection, and wherein adaptive pivots are tracked independent of final outcome success enabling measurement of strategic flexibility as separate capability from result achievement.

28. The system of claim 26 wherein the biases recognized accumulator tracks three bias types comprising confirmation bias detected through pattern matching on repeated failed strategies, anchoring bias detected through unchanged initial estimates despite contradictory evidence, and halo effect detected through positive actor attribute overgeneralization, with bias events triggered through user acknowledgment OR system inference mechanisms.

29. The system of claim 26 wherein the dashboard visualization provides equal visual prominence to metacognitive awareness metrics (biases recognized) and task performance metrics (successful ROI and success rate), reinforcing importance of self-monitoring skills in communication training through UI design emphasizing metacognitive development.

30. The system of claim 26 further comprising an analytics dashboard displaying skill progression charts for observation, adaptation, and self-awareness skills tracked across interaction timeline showing baseline values, effective values after confounder adjustments, and trend lines indicating improvement or degradation trajectories, with milestone markers for significant capability achievements.

31. The system of claim 30 further comprising a prediction accuracy analysis panel displaying time-series trends for prediction errors across strategic outcomes with mean absolute error calculations, trend analysis indicating cumulative learning effects, calibration plots comparing predicted probability distributions to actual outcome frequencies, and systematic bias pattern identification highlighting actor-specific or strategy-specific estimation errors with recalibration recommendations.